

Test 17: Connes self-duality ??1/?

Verdict: PASS | Generated: 2026-09-02 15:07:36 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

EXACT: isospectrality under ? ? 1/? to machine precision.

METHOD: EXACT: isospectrality under ? ? 1/? to machine precision.

Test 17 HARDCORE L=24: zero_modes(?=1/2)=4, spectral defects 1.42e-15/1.66e-15, duality defect 3.62e-01.

Test 17 HARDCORE L=48: zero_modes(?=1/2)=4, spectral defects 2.90e-15/2.41e-15, duality defect 3.65e-01.

Test 17 HARDCORE L=96: zero_modes(?=1/2)=4, spectral defects 3.96e-15/3.24e-15, duality defect 3.66e-01.

Test 17b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

Result

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 17 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[14:40:55.756] [+3171.039s] build_ab_cloud_hamiltonian: 24x24 (N=576), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[14:40:59.296] [+3174.578s] build_ab_cloud_hamiltonian: 24x24 (N=576), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[14:41:00.067] [+3175.349s] build_ab_cloud_hamiltonian: 24x24 (N=576), ?=0.3333, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[14:41:00.859] [+3176.141s] build_ab_cloud_hamiltonian: 24x24 (N=576), ?=0.5000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[14:41:01.457] [+3176.739s] build_ab_cloud_hamiltonian: 24x24 (N=576), ?=2.0000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[14:41:02.710] [+3177.992s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[14:41:08.135] [+3183.417s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[14:41:13.525] [+3188.807s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.3333, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[14:41:19.476] [+3194.758s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=0.5000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[14:41:20.433] [+3195.715s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ?=2.0000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[14:41:32.373] [+3207.655s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[14:46:45.944] [+3521.226s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[14:52:04.211] [+3839.493s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.3333, t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[14:57:25.070] [+4160.352s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.5000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

[14:57:36.057] [+4171.339s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=2.0000, t=1.00, W=4.00, open, 0 vortices, model=:monumental

CONSOLE CAPTURE

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TEST 17 (pass 2, HARDCORE): Connes self-duality ? cross-size audit

Zero-mode invariant + spectral symmetries on the size ladder 24 ? ab_nx

??

L=24: zero_modes(?=1/2)=4 | spec.defect(1/2)=1.42e-15, (1/3)=1.66e-15 | duality=3.62e-01

L=48: zero_modes(?=1/2)=4 | spec.defect(1/2)=2.90e-15, (1/3)=2.41e-15 | duality=3.65e-01

L=96: zero_modes(?=1/2)=4 | spec.defect(1/2)=3.96e-15, (1/3)=3.24e-15 | duality=3.66e-01

At ?=1/3 the spectrum stays gapped at E=0 on these tori (base observation).

??

zero_modes(1/2)=4 ?L PASS ? topological invariant holds on the whole ladder

E??E defect < 1e-10 ?L PASS ? ?=1/2 and ?=1/3, generic bipartite symmetry

Test 17b [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS